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Pharmacomicrobiomics in blood cancers: How the gut microbiome and its metabolites shape drug efficacy and toxicity

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ABSTRACT

Clinical management of hematologic cancers is frequently complicated by marked, unpredictable inter-patient variation in both therapeutic benefit and adverse effects. Beyond host genetics, accumulating mechanistic and translational data implicate the gut microbiota and its small-molecule metabolome as active modifiers of drug chemistry, host immune tone, and clinical outcomes. In this review we synthesize three recurring, actionable findings: (1) discrete microbial enzymes, most prominently bacterial β-glucuronidases, repeatedly re-activate hepatic drug-glucuronides in the gut and amplify local gastrointestinal toxicity (e.g., irinotecan and mycophenolate), a pathway successfully targeted in preclinical and translational studies. (2) loss of anaerobic short-chain fatty acid production, especially butyrate, is a recurrent, causal mediator of mucosal injury and aggravated graft-versus-host disease after allogeneic hematopoietic cell transplantation. (3) reduced gut microbial diversity and the depletion of key functional taxa predict worse transplant outcomes (including mortality), arguing that baseline ecosystem state is a prognostic biomarker that merits routine consideration in trial design. Building on these syntheses, we propose a compact clinical framework for hematology trials and practice: (A) baseline ecosystem phenotyping (shotgun metagenome + targeted metabolite panel including butyrate and measured bacterial β-glucuronidase activity), (B) risk-stratified interventions (enzyme-targeted inhibitors, defined consortia, or metabolite replacement for high-risk patients), and (C) embedded holo-omic endpoints to confirm on-target biochemical modulation and link molecular change to clinical benefit. Together, these elements move Pharmacomicrobiomics beyond descriptive cataloging toward mechanistic stratification and testable

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interventions that can reduce unexplained variability in drug response and toxicity for patients with leukemia, lymphoma and myeloma.

1. Introduction

Precision oncology has transformed hematologic malignancy care by linking tumor genomics and host pharmacogenetics to therapeutic choice, yet clinically important interpatient variation in drug efficacy and toxicity remains largely unexplained in leukemia, lymphoma, and myeloma cohorts (Moriyama et al., 2016). Classical host pharmacogenomic markers such as TPMT variants and more recently discovered NUDT15 alleles reduce but do not eliminate thiopurine toxicity and together account for only a fraction of observed adverse events, illustrating a persistent unaccounted biological variance that constrains truly individualized dosing (Meggitt et al., 2006; Schaeffeler et al., 2019). Beyond single gene effects, population studies in hematology demonstrate that genomic predictors frequently fail to predict complex outcomes such as chemotherapy tolerance, infection risk during neutropenia, and immunotherapy responsiveness, arguing that additional, nonhost genomic axes of variation must be integrated into precision models (Peled et al., 2020). Emerging mechanistic and translational data increasingly treat the gut microbiome as a pharmacologically relevant contributor that may chemically and functionally modulate systemic drug exposure, therapeutic activity, and toxicity; however, robust causal demonstration in humans remains limited and often derives from complementary preclinical models (Haiser et al., 2014; Wallace et al., 2010). Microbial genomes encode enzymes that can biotransform xenobiotics; canonical, well-studied proof-of-principle examples are the Eggerthella lenta–mediated reduction of digoxin and bacterial β-glucuronidase reactivation of irinotecan metabolites. These serve as mechanistic paradigms demonstrating how microbial enzymes can change drug potency, enterohepatic recycling, and local mucosal injury. Crucially for this review, the same enzymatic logics apply to hematology-relevant agents (for example mycophenolate, tacrolimus, and methotrexate), which can undergo gut microbial deconjugation or direct metabolism with direct consequences for transplant and systemic hematology practice (Haiser et al., 2014; Wallace et al., 2010). In parallel, commensal-derived small molecules act as endocrine-like mediators that traverse the gut barrier and engage host receptors to reprogram immune effectors, modulate hematopoietic niches, and alter the tumor microenvironment in ways that change drug response (Mager et al., 2020; Zhang et al., 2022).

Blood-cancer care is a particularly informative clinical setting for studying microbiome-dependent pharmacology because intensive cytoreduction, prolonged neutropenia, broad-spectrum antibiotic exposure, and allogeneic transplantation repeatedly perturb microbial ecosystems, with resulting compositional and functional shifts that are reproducibly associated with mortality, graft-versus-host disease severity, and infectious complications (Mathewson et al., 2016; Peled et al., 2020). Mechanistic animal and human translational work shows that loss of key microbial metabolites such as butyrate exacerbates mucosal damage and immune dysregulation after conditioning, and that microbiota-driven modulation of marrow iron handling and hematopoietic stem cell fate can influence recovery and treatment tolerance (Mathewson et al., 2016; Zhang et al., 2022).

We submit that Pharmacomicrobiomics 2.0 should be framed as an evolution from descriptive ecological catalogs toward a mechanistic and translational discipline that aims to establish causal links where possible and to develop therapeutic approaches guided by rigorous experimental and clinical validation (Nayak et al., 2021; Wallace et al., 2015). Historical milestones have catalyzed this shift. Early mechanistic proofs that commensal bacteria can directly inactivate drugs (for example, digoxin inactivation by Eggerthella lenta) established the concept of microbial drug metabolism. Seminal enzyme-targeting work demonstrated that inhibiting bacterial β-glucuronidases mitigates chemotherapy-related gut toxicity, proof that microbial enzymology can be a therapeutic target. In hematology, large multi-center cohorts linked peri-transplant loss of gut diversity to higher transplant-related mortality and worse acute GVHD outcomes, providing the first reproducible clinical outcome signal. Finally, the field progressed from association to intervention with responder-derived fecal microbiota transplant (FMT) trials that restored checkpoint sensitivity in some refractory melanoma patients, and with the first randomized studies of defined, regulatory-grade bacterial consortia (for example VE303 and SER-109) that demonstrate clinical activity and the feasibility of standardized microbial therapeutics (Davar et al., 2021; Feuerstadt et al., 2022; Haiser et al., 2014; Louie et al., 2023). This second generation agenda prioritizes genome-resolved metagenomics, metatranscriptomics, and untargeted metabolomics paired with gnotobiotic and humanized models and early phase human intervention trials to move from association to causation and to validate microbiome-targeted interventions that increase therapeutic index (Wallace et al., 2015, 2010).

The present review synthesizes the most recent human and experimental studies documenting three interlocking mechanisms by which the gut ecosystem shapes therapy in hematologic malignancies. These are: (i) direct microbial drug metabolism that alters drug pharmacokinetics and bioactivity (for example, bacterial β-glucuronidase–mediated reactivation of drug glucuronides); (ii) metabolite-mediated modulation of systemic and tumor immunity (for example, short-chain fatty acids modulating mucosal integrity and immune tone); and (iii) marrow–microbiome cross-talk that influences hematopoietic recovery and iron handling in bone marrow (Guo et al., 2019; Nayak et al., 2021; Zhang et al., 2022). Our novelty lies in emphasizing microbial enzymology as a determinative axis of drug fate, integrating holo-omic discovery with causal experimental pipelines, and outlining translational strategies for microbiome-assisted therapy including selective enzyme inhibitors, metabolite replacement, and standardized microbial consortia to reduce the unexplained variance that remains after host genomic stratification (Mager et al., 2020; Wallace et al., 2010).

By recognizing the gut ecosystem as an active contributor to drug metabolism and response and by articulating a pragmatic research and translational roadmap, we contend that inclusion of microbiome variables is now essential for advancing precision oncology in leukemia, lymphoma, and myeloma (Mager et al., 2020; Wallace et al., 2010). An overview of the principal microbiome-mediated mechanisms that alter drug pharmacology is summarized in Table 1.

2. Microbiota–BM crosstalk: systemic signals that shape hematopoiesis

2.1. Homeostatic and stress-responsive regulation of hematopoietic stem cells by gut-derived molecules

The intestinal microbial community provides systemic cues that can influence bone-marrow function: bacterially derived molecules that enter the bloodstream may be interpreted by marrow niches and have been shown in experimental systems to bias hematopoietic stem cell (HSC) self-renewal and downstream lineage allocation (Liu et al., 2024; Zhang et al., 2022). Mechanistic studies reveal a microbiota → marrow macrophage → iron-handling cascade that biases HSC fate under regenerative pressure: perturbing the microbiota shifts macrophage iron recycling and, in some experimental settings, increases HSC self-renewal while reducing differentiation; the direction of this effect depends on the specific perturbation and contextual signals (Zhang et al., 2022). Converging lines of evidence show that microbially produced

Table 1
Mechanisms by which the gut microbiome shapes drug efficacy and toxicity.

Mechanism	Short description	Ref.
Microbial enzymatic drug biotransformation (β-glucuronidase)	Bacterial β-glucuronidases reactivate drug-glucuronides in the gut, increasing local toxicity (e.g., irinotecan, mycophenolate).	(Bhatt et al., 2020; Wallace et al., 2010)
Direct drug inactivation by specific taxa (e.g., Eggerthella lenta)	Some commensals enzymatically reduce or inactivate drugs. Digoxin inactivation by Eggerthella lenta is a canonical proof-of-principle example that demonstrates direct microbial drug inactivation; analogous mechanisms (e.g., enzymatic deconjugation or metabolism) have been implicated in hematology-relevant compounds such as mycophenolate and tacrolimus.	(Haiser et al., 2014)
Microbial metabolite modulation of mucosal injury & GVHD	Loss of SCFAs (notably butyrate) worsens mucosal damage and graft-versus-host disease after allo-HCT.	(Mathewson et al., 2016)
Microbiome → bone-marrow (iron-handling) cross-talk	Microbial signals alter marrow macrophage iron recycling and thereby HSC fate decisions under stress.	(Zhang et al., 2022)
Microbiome-driven immune priming / adjuvancy	Translocation or antigen-sampling of commensals can boost antitumor T-cell priming (relevant to alkylators & immunotherapy).	(Daillère et al., 2016; Gopalakrishnan et al., 2018)

short-chain fatty acids (e.g., butyrate) and related metabolites transit into circulation, penetrate marrow microenvironments, and reprogram progenitors and niche populations, altering myeloid maturation and macrophage phagocytic functions that secondarily affect erythrophagocytosis and iron homeostasis (Liu et al., 2024; Zhang et al., 2022). Acute disruption of the mucosa, particularly signals from Gram-negative commensals, can trigger innate cascades that expand HSC/progenitor compartments and mobilize them to sites of inflammation; experimental data implicate Bacteroides-linked cues and GM-CSF dependent mechanisms in this response (Sezaki et al., 2022). At the cellular interface, HSCs and progenitors display pattern-recognition receptors (notably TLRs) and can directly detect microbial molecules; for instance, systemic LPS activates HSCs through TLR4–TRIF signaling causing transient proliferation but, if prolonged, diminished long-term repopulating fitness (Nagai et al., 2006; Takizawa et al., 2017). Together, these data support a two-mode model: basal, low-amplitude microbial metabolites and MAMPs homeostatically bias lineage output and niche behavior, whereas acute or sustained microbial inflammation drives emergency myelopoiesis and risks HSC exhaustion (Liu et al., 2024; Takizawa et al., 2017). Clinically relevant perturbations, broad-spectrum antibiotics, major dietary shifts in fiber, or the severe dysbioses that follow transplantation, modify microbial metabolite landscapes and thereby reshape marrow output and immune reconstitution; such shifts associate with heightened infection risk, worsened GVHD, and may even influence clonal dynamics in bone-marrow disorders (Le Bastard et al., 2021; Romick-Rosendale et al., 2018). As illustrated in Fig. 1, gut-derived metabolites and microbial components form an integrated gut–liver–bone marrow axis that can reprogram hematopoietic niches and influence drug fate and antitumor immunity.

2.2. Pattern recognition, direct sensing and mechanistic receptors

A gut ecosystem characterized by beneficial commensals establishes a systemic immune landscape that enhances tumor surveillance through improved dendritic cell activation, more effective T-cell priming, and increased infiltration of effector lymphocytes into malignant tissues (Mager et al., 2020; Perl et al., 2025). Across several human cohorts, greater intestinal microbial richness, or the presence of specific favorable taxa, has been correlated with improved clinical outcomes following immune checkpoint inhibitor therapy (Allard et al., 2017; Mager et al., 2020; Wang et al., 2025). Intervention experiments in germ-free and humanized mice, including fecal microbiota transfer from human responders, have transferred enhanced antitumor T-cell activity and improved checkpoint responsiveness to recipients; these data

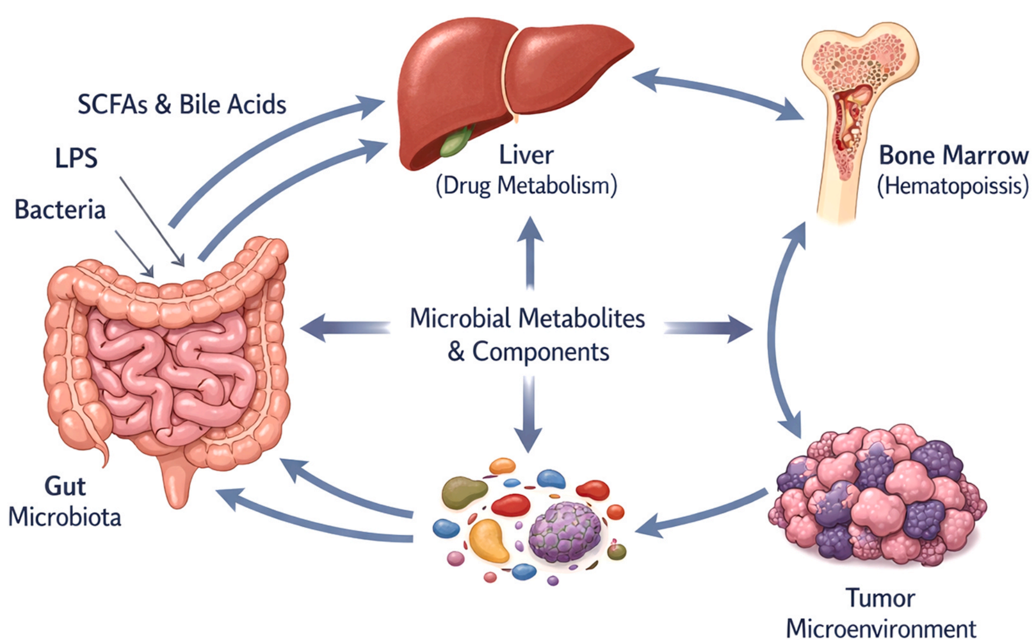

Fig. 1. The central gut–bone marrow–lymphoid axis. Gut microbial communities produce metabolites and microbe-associated molecular patterns (e.g., short-chain fatty acids, bile acids, LPS) that travel to the liver and bone marrow, modulate drug metabolism and hematopoietic niche function, and reshape the tumor microenvironment and systemic immunity. This integrated network highlights paths by which microbial chemistry can alter hematopoiesis, drug pharmacokinetics, and antitumor immune responses.

provide strong preclinical support for a causal role of resident microbial communities and their metabolites, while human interventional data remain limited and variable (Bird, 2020; Gopalakrishnan et al., 2018; Sun et al., 2025). Mechanistic work attributes response-associated microbes to enhancement of antigen-presenting cell maturation, potentiation of type-I interferon pathways, and promotion of CD8⁺ T-cell tumor homing; in parallel, bacterially produced small molecules (for example, SCFAs) tilt the effector/regulatory T-cell axis and reshape myeloid suppressor programs that normally limit immunotherapy (Peled et al., 2020; Zhu et al., 2023). Early interventional human trials in solid tumours (notably responder-derived FMT in PD-1-refractory melanoma) have produced objective clinical responses in subsets of patients and established proof-of-principle that microbiome modulation can alter intratumoral immunity (Baruch et al., 2021; Davar et al., 2021). By contrast, hematology-specific interventional data remain limited: most hematologic evidence is currently prospective or retrospective cohort work that links baseline microbiome/metabolome state to CAR-T and HSCT outcomes (Artacho et al., 2021; Reikvam et al., 2023; Smith et al., 2022), while a small number of early case series and pilot FMT/consortium studies report feasibility and potential benefit in transplant recipients (Gray and DeFilipp, 2023; Weber et al., 2024). Parallel hematology-specific longitudinal cohorts and metabolic profiling studies (including the multicenter CAR-T cohorts and single-center prospective metabolome studies) show that pre-infusion microbiome/metabolome states predict CAR-T toxicity and response, supporting a translational pathway for metabolome-guided patient stratification in cellular therapies (Hu et al., 2022; Smith et al., 2022; Uribe-Herranz et al., 2025). Although the most decisive interventional evidence comes from solid-tumor studies (especially melanoma), direct trials in hematologic cancers remain scarce; consequently, translating those microbial strategies into blood-cancer immunotherapy or adoptive-cell therapy protocols requires careful, disease-specific validation rather than straightforward extrapolation (Guevara-Ramírez et al., 2023; Le Bastard et al., 2021).

3. Drug–microbe interactions in hematologic therapeutics

3.1. Antimetabolite–microbiome interfaces: mechanisms, sites of action, and clinical consequences

Antimetabolites are central to many treatment regimens for blood cancers, and they engage in complex, two-way interactions with the host microbiota that can alter both therapeutic benefit and adverse-event profiles. Methotrexate (MTX) exemplifies this class: a body of in vitro, ex vivo and translational work shows that MTX perturbs gut bacterial growth and community composition, and that many gut bacteria harbour folate-pathway genes (including genes related to folylpolyglutamate metabolism), providing a mechanistic basis for microbe–MTX interactions (Artacho et al., 2021; Nayak et al., 2021). Mechanistic and review evidence point to the gut lumen and mucosa-associated bacterial communities as plausible sites of microbial MTX modification, consistent with intracellular enzymology that would require bacterial contact with luminal or adherent drug pools; however, direct detection of bacterially-derived MTX-polyglutamate species in human systemic compartments is still limited (Villa Soto et al., 2025; Yan et al., 2021). Because polyglutamylation-type reactions require bacterial intracellular enzymes and substrate uptake, luminal or mucosa-attached taxa that physically encounter orally delivered (or biliary-excreted) MTX are biologically the most plausible mediators of direct chemical alteration; this is corroborated by in vitro strain screens showing that isolated gut bacteria can consume or chemically convert MTX (Nayak et al., 2021; Villa Soto et al., 2025; Yan et al., 2021). Nevertheless, microbial activity at the gut surface can affect systemic MTX exposure via two clinically relevant pathways: (i) local changes in mucosal retention or metabolism that alter the fraction of parent drug absorbed into the circulation, and (ii) barrier disruption (e.g., post-conditioning mucositis) that permits translocation of microbes, microbial enzymes, or chemically modified drug species into systemic tissues and lymphoid organs, mechanisms that are supported by animal models linking mucosal injury, bacterial translocation, and systemic pharmacologic or immunologic effects (Wilson and Nicholson, 2017; Zhou et al., 2018). Empirical detection of bacterially-derived MTX-polyglutamate species in human stool, mucosal biopsies, or plasma remains scarce; determining where modification predominates will therefore require focused metatranscriptomic/metaproteomic profiling of mucosa-associated microbes together with targeted metabolomics that quantifies MTX and its polyglutamated derivatives across stool, mucosa, and blood (Yan et al., 2021).

Both preclinical and human data indicate that MTX use alters gut community structure in a dose-dependent fashion, and that interpatient variation in pretreatment microbiota correlates with differences in clinical response and the risk of mucosal side effects (Artacho et al., 2021; Nayak et al., 2021). In mechanistic terms, microbial processes deserve special emphasis: a) Enzymatic capacity (demonstrated vs hypothesised): Demonstrated: multiple bacterial genomes encode folate-pathway enzymes and in vitro/ex vivo experiments show bacteria can modify folates and deplete MTX (Artacho et al., 2021; Nayak et al., 2021). Hypothesis: that these bacterial enzymes produce clinically meaningful MTX-polyglutamate (MTX-PG) pools in vivo that alter host intracellular drug polyglutamylation remains to be conclusively shown, the existing data are largely in vitro/ex vivo. Discriminating these alternatives requires assays that quantify MTX and MTX-PG species across stool, mucosa and plasma alongside metatranscriptomic/proteomic evidence of active microbial FPGS-like expression (de Crécy-Lagard et al., 2007; El Fadili et al., 2003), b) Barrier and inflammatory modulation: Robust preclinical and translational studies show that microbiome composition controls mucosal barrier integrity and inflammatory tone in the gut (and liver), thereby influencing two major MTX toxicities, intestinal mucositis and hepatotoxicity; antibiotic depletion or specific taxon enrichment changes the severity of MTX-induced mucositis in animal models and associates with differing clinical phenotypes in humans (Letertre et al., 2020; Zhou et al., 2018), c) Niche-specific contributors (hypothesis): genomic surveys indicate that certain skin and mucosal commensals, for example Cutibacterium acnes, harbour folate-pathway gene clusters, including folC and FPGS-like homologues, which could enable local folate/antifolate chemistry. However, it remains unproven whether these taxa actually mediate MTX chemical modification or influence clinical outcomes in vivo (oral mucosa, small intestine, skin); accordingly, this should be framed as a testable hypothesis rather than an established mechanism. Targeted in-situ metabolomics combined with spatial transcriptomics of mucosal niches will be required to evaluate this possibility (Becker et al., 2023; Mazhar et al., 2023; Yang et al., 2024; Yu et al., 2024). Taken together, these strands argue for a tripartite pharmacology model in which host biology, drug properties, and microbial enzymology jointly determine MTX efficacy and toxicity, a framework that explains interindividual variability not accounted for by host genetics alone (Chen et al., 2024; de Crécy-Lagard et al., 2007).

Thiopurines (azathioprine and 6-mercaptopurine) further illustrate microbiome-drug interplay: while host TPMT and NUDT15 genotype remain major determinants of toxicity, many bacterial genomes encode SAM-dependent methyltransferases homologous to eukaryotic TPMT, and recent studies show commensal organisms can catabolize azathioprine/6-MP, reducing drug bioavailability and, in some models, promoting therapeutic failure, whereas microbiota depletion can worsen thiopurine-related intestinal toxicity. These observations imply that microbial TPMT-like activity and other bacterial catabolic routes act alongside host genetics to shape systemic thiopurine exposure, making microbial gene families potential predictive biomarkers or targets for adjunctive microbiome interventions (Asadov et al., 2017; Dean L, 2012; Jantararoungtong et al., 2021; Lazarević et al., 2022; Scheuermann et al., 2003; Yan et al., 2023).

3.2. Alkylators as immune adjuvants

Rather than acting solely as a DNA-alkylating cytotoxin, cyclophosphamide functions also as an immune-modulating agent whose capacity to reprogram antitumor immunity is intimately linked to the intestinal microbiota (Daillère et al., 2016; Scurr et al., 2017). Seminal mouse studies demonstrated that select Gram-positive commensals, including Enterococcus hirae and Barnesiella intestinihominis, migrate to mesenteric lymphoid tissues following cyclophosphamide; in these models this migration appears to be necessary for driving peripheral Th1/Th17 expansion and improved intratumoral CD8:Treg ratios that correlate with prolonged tumor control, but whether the same mechanisms are necessary for clinical activity in humans remains to be established (Daillère et al., 2016; Scurr et al., 2017). Mechanistic work indicates that CTX-induced mucosal injury selectively permits commensal antigens to reach lymphoid compartments, where they reprogram antigen-presenting cells and bias T-cell differentiation toward IFN-γ and IL-17 producing states that cooperate with cytotoxic damage to enhance malignant cell clearance (Daillère et al., 2016). Crucially, CTX's balance of immunostimulatory versus overtly cytotoxic actions varies with dose. Thus, metronomic or low-dose CTX tends to produce transient Treg impairment and augment antitumor T-cell activity, while high-dose regimens cause profound cytotoxicity, extensive mucosal damage and generalized immunosuppression, a distinction that determines whether microbial signals support immune priming or pathologic inflammation. In sum, dose governs outcome: when CTX preserves controlled mucosal sampling of commensals it can act as an adjuvant to antitumor immunity, but when dosing produces extensive barrier breakdown the same microbiome-related processes may instead precipitate infection or harmful inflammation (Daillère et al., 2016; Kachur et al., 2023; Scurr et al., 2017).

3.3. Microbial metabolites as co-modifiers of IMiD immunobiology

Lenalidomide and pomalidomide represent the prototypical immunomodulatory imide drugs (IMiDs), whose principal antitumor mechanism involves binding to cereblon and altering CRL4ˆCRBN ubiquitin-ligase substrate selection in malignant plasma cells and immune effectors (Wang et al., 2024). However, clinical correlations and mechanistic studies indicate that host microbial metabolites, most notably SCFAs such as butyrate, can intersect with IMiD-driven immune reprogramming at several mechanistic nodes (O'Riordan et al., 2022; Silva et al., 2020). Butyrate acts as a strong epigenetic effector (via histone-deacetylase inhibition and related pathways) that shapes T-cell polarization and myeloid biology; measured increases in colonic and circulating butyrate correlate with expanded regulatory T-cell programs and altered antigen-presentation/cytokine profiles that could materially modify the way IMiDs reconfigure the tumor milieu (Duan et al., 2025; O'Riordan et al., 2022; Silva et al., 2020). In observational multiple-myeloma series, patients with elevated stool butyrate levels and plant-forward diets that favor butyrogenesis demonstrated higher rates of sustained minimal-residual-disease negativity while on lenalidomide maintenance, an associative signal that supports the hypothesis of SCFA–IMiD synergy but requires prospective interventional testing (Jasiński et al., 2021; Shah et al., 2022; Wang et al., 2024). Mechanistically, butyrate's HDAC-inhibitory and signaling properties can strengthen T-cell effector differentiation and modify antigen-presenting cell function; such complementary actions could magnify cereblon-dependent immune effects of IMiDs without altering direct cereblon binding, creating a testable framework where dietary measures or microbiome-directed enrichment of butyrogenic taxa serve as logical adjuvants to IMiD regimens (Duan et al., 2025; Jasiński et al., 2021; Silva et al., 2020). Because IMiDs are foundational in maintenance and salvage strategies for plasma-cell malignancies, embedding microbiome and metabolome endpoints into upcoming clinical trials, including dietary, prebiotic, and defined-consortium interventions, would efficiently test whether augmenting butyrate biology yields reproducible clinical enhancement of IMiD activity (Shah et al., 2022).

3.4. From gut ecology to checkpoint efficacy

Over the past ten years, human and preclinical evidence has identified the intestinal microbiota as an influential factor associated with anti–PD-1/PD-L1 therapy outcomes in solid tumours; mechanistic principles from these studies (microbial metabolites, antigenic priming, and myeloid conditioning) are potentially pertinent to hematologic cancers treated with checkpoint blockade, although confirmatory clinical trials in hematology are required (Gopalakrishnan et al., 2018; Peled et al., 2020; Routy et al., 2018). Notably, clinical cohorts have associated responder status with enrichment of taxa including Akkermansia muciniphila, Faecalibacterium spp., and Bifidobacteria, and causality was reinforced when fecal transplants from responding patients conferred enhanced checkpoint sensitivity to germ-free or antibiotic-treated mice (Baruch et al., 2021; Gopalakrishnan et al., 2018; Routy et al., 2018). Mechanistic frameworks invoked to explain these associations span improved antigen-presenting cell maturation and cross-reactive T-cell expansion (including antigen mimicry), alterations in bile-acid and tryptophan catabolite pathways that steer T-cell trafficking/function, and SCFA-mediated shifts in effector-versus-regulatory T-cell balance (Ansaldo et al., 2019; Dora et al., 2023; Mager et al., 2020). Conversely, particular community configurations (for example Bacteroidales-dominant profiles or other adverse signatures) have correlated with primary checkpoint resistance or heightened immune-related toxicity, underscoring that community structure and functional output, not single taxa in isolation, usually determine the clinical effect (Dora et al., 2023; Gharaibeh and Jobin, 2019; York, 2018). Applying these insights to blood cancers demands careful attention to disease-specific biology and the unique treatment exposures that shape host–microbiome interactions (Peled et al., 2020; Zhang et al., 2023). For instance, PD-1-sensitive entities (classical Hodgkin lymphoma and certain primary mediastinal B-cell lymphomas) are now the subject of translational cohorts, including the MicroLinf initiative, that seek to determine whether the 'responder' microbiome signatures discovered in solid tumours predict checkpoint benefit in lymphoma (Casadei et al., 2024; Derosa et al., 2022; Taur et al., 2014). Importantly, hematologic patients frequently carry exposures (intensive cytotoxic regimens, prolonged neutropenia, prior transplantation and peri-transplant antibiotics) that profoundly remodel mucosal communities and immune reconstitution, factors that can amplify, dampen, or confound microbiome-checkpoint associations extrapolated from solid-tumour work (Henig et al., 2021; Zhang et al., 2023). Collectively, the data justify a translational pathway in hematologic immunotherapy: incorporate baseline and longitudinal microbiome/metabolome profiling into trials, and rigorously test targeted manipulations (e.g., rational FMT, defined consortia, dietary or metabolite interventions) to assess whether altering community composition or function can raise response rates or mitigate immune-related toxicities (Baruch et al., 2021; Henig et al., 2021; Louie et al., 2023; Sun et al., 2023).

3.5. Peri-transplant microbial pharmacology

Allogeneic hematopoietic cell transplantation (allo-HCT) exemplifies a clinical setting where gut microbial ecology, microbial drug transformations, and immune sequelae intersect and jointly determine patient trajectories (Henig et al., 2021; Peled et al., 2020). Large multicenter observational series have consistently associated peri-transplant reductions in gut microbial alpha-diversity with worse outcomes, notably higher transplant-related mortality and more severe aGVHD, though these cohort associations do not by themselves constitute validated prognostic assays without formal modelling and external validation (Le Bastard et al., 2021; Peled et al., 2020; Steyerberg et al., 2010; Taur et al., 2014). Preparative conditioning (high-dose cytoreduction ± irradiation) produces rapid ecological collapse in the intestine, often marked by loss of butyrate-producing commensals, overgrowth (taxonomic domination) of opportunistic Proteobacteria, and reduced community resilience, alterations that sensitize the mucosa to injury and amplify inflammatory pathways predisposing to GVHD (Henig et al., 2021; Soleimani Samarkhazan et al., 2025; Yue et al., 2024). These ecological shifts have pharmacologic consequences: drugs routinely given around transplant, including immunosuppressants and antimicrobial prophylactics, undergo microbiome-dependent biotransformations that can materially change their enterohepatic cycling and systemic exposure (Degraeve et al., 2023; Drevland et al., 2025). Prospective, longitudinal metabolome and microbiome studies in allogeneic HSCT cohorts now reinforce these mechanistic links: pre-transplant systemic metabolic profiles and peri-transplant fecal metabolome shifts predict acute GVHD and transplant-related mortality in prospective cohorts (Artacho et al., 2021; Kim et al., 2025; Reikvam et al., 2023). Concretely, mycophenolate-derived glucuronides (MPAG) can be enzymatically deconjugated by bacterial β-glucuronidases in the gut, restoring active mycophenolic acid to the lumen and circulation; this process correlates with increased enterohepatic recirculation and gastrointestinal toxicity in preclinical and translational studies (Drevland et al., 2025; Dukaew et al., 2023). Likewise, calcineurin inhibitors (e.g., tacrolimus) and other agent classes can be chemically altered or have their pharmacokinetics influenced by gut microbes: both cohort data and mechanistic work implicate particular taxa (for example Faecalibacterium-linked activities) in modulating tacrolimus metabolism and exposure after transplantation (Degraeve et al., 2023; Guo et al., 2019). Because drug concentrations calibrate the delicate trade-off between graft-versus-leukemia efficacy, GVHD risk, and infection susceptibility, microbiome-mediated shifts in drug handling are of direct clinical importance in allo-HCT management (Henig et al., 2021; Saqr et al., 2025). Moreover, standard peri-transplant practices, prolonged broad-spectrum antibiotics, gut decontamination strategies, and extended immunosuppression, further remodel the microbiome and hence secondarily alter the metabolic fate of prophylactic and therapeutic drugs that would otherwise display predictable pharmacology (Dukaew et al., 2023; Kim et al., 2025; Peled et al., 2020).

Table 2 summarizes representative, clinically relevant microbe—drug interactions discussed in Section 3 and highlights taxa and mechanisms that can alter pharmacokinetics and toxicity in hematologic patients.

Fig. 2 provides representative molecular examples, bacterial folate-pathway interactions with MTX, enzymatic antibiotic degradation, and microbe-derived immune modulators, that illustrate the drug-microbe mechanisms discussed throughout Section 3.

4. Translational tools: diagnostics, therapeutics, and engineering the metabolome

4.1. Microbiome- and metabolome-based clinical biomarkers

Pre-treatment gut community structure and the small-molecule profile it produces have been consistently associated with clinical outcomes (for example, lower α-diversity around engraftment correlates with higher transplant-related mortality and worse aGVHD). These signals currently represent candidate prognostic biomarkers or cohort-level predictors rather than clinically validated assays; moving them toward clinical use will require explicit model reporting (discrimination and calibration metrics) and external validation in independent cohorts (Collins et al., 2015; Steyerberg et al., 2010). For clarity and reproducibility, studies proposing microbiome-based predictors should follow established reporting and validation frameworks (e.g., TRIPOD / REMARK) and provide explicit performance metrics, e.g., discrimination (AUC/C-statistic), calibration (calibration plot, calibration slope), and, when possible, external validation in independent cohorts, and should be evaluated for risk of bias (PROBAST). These steps help distinguish preliminary associations or pilot predictive models from assays ready for clinical deployment (Collins et al., 2015; Sauerbrei et al., 2018; Steyerberg et al., 2010). Moving past species lists alone, both targeted and global metabolomic assays of feces and tissue have singled out microbial-derived compounds, for example, short-chain fatty acids (butyrate, propionate, whose loss is linked to mucosal damage and a higher incidence of severe gastrointestinal GVHD, indicating that metabolite signatures can serve as mechanistic, actionable biomarkers rather than inert correlates (Andersen et al., 2025; Lou et al., 2024; Mathewson et al., 2016; Song et al., 2024; Yazdandoust et al., 2023). Importantly, several hematology-focused, prospective studies have integrated pre-transplant metabolomics with serial fecal profiling and linked specific metabolite patterns (for example taurine/tryptophan pathway perturbations and SCFA depletion) to subsequent aGVHD and early transplant mortality (Artacho et al., 2021; Kim et al., 2025; Reikvam et al., 2023). In the setting of adoptive cellular therapies, multinational cohorts have reported that pre-infusion gut microbial patterns and dynamic metabolomic changes during therapy associate with both response and adverse outcomes after CAR-T infusion, for instance, with prediction of cytokine-release syndrome intensity and probabilities of sustained remission, implying that combined microbiome–metabolome predictive models may stratify patients before and during cellular therapy (Kim et al., 2025; Perna et al., 2024; Smith et al., 2022; Zhang and Xie, 2024).

Table 2
Key microbe → drug interactions relevant to hematologic care.

Drug / drug class	Microbial mechanism (what bacteria do)	Representative taxa / enzyme	Clinical consequence (in hematology)	Ref.
Digoxin	Microbial reduction / inactivation of drug	*Eggerthella lenta,* cardiac-glycoside reductase operon	Reduced digoxin activity; serves as a canonical proof-of-principle of direct microbial drug inactivation, mechanistically analogous to microbiome-driven modification of hematology-relevant agents (see mycophenolate and tacrolimus rows).	(Haiser et al., 2014)
Methotrexate (MTX) / antifolates	Direct bacterial DHFR / folate-pathway interactions; dose-dependent community shifts	Diverse gut bacteria with folate-pathway enzymes; mucosa-associated taxa	Altered MTX bioavailability, mucositis risk; pretreatment microbiome associates with response.	(Nayak et al., 2021; Yan et al., 2023)
Thiopurines (azathioprine / 6-MP)	Bacterial TPMT-like methyltransferases; bacterial catabolism reduces drug availability	Various commensals (e.g., *Blautia* spp.* implicated in IBD*)	Possible reduced systemic exposure → therapy failure or changed toxicity profile.	(Yan et al., 2023)
Mycophenolate (MMF / MPA)	Glucuronide deconjugation by bacterial β-glucuronidases → reactivation and enterohepatic recirculation	GUS-expressing taxa (e.g., *Bacteroides* spp.)	Increased GI toxicity and variable systemic exposure; antibiotic modulation affects toxicity.	(Drevland et al., 2025)
Tacrolimus	Microbial conversion to less-potent metabolites	Commensal bacteria demonstrated to metabolize tacrolimus	May contribute to variable tacrolimus levels after oral dosing (pharmacokinetic variability).	(Guo et al., 2019)

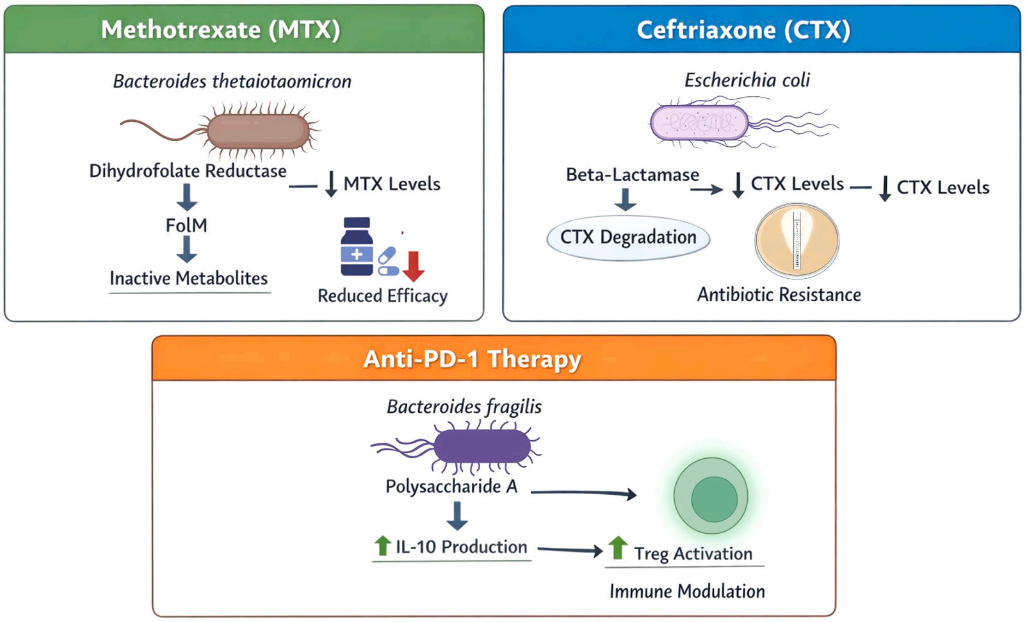

Fig. 2. Representative molecular mechanisms by which gut microbes modify drugs and immune therapies. (A) Bacteroides and related taxa possess folate-pathway enzymes that can alter methotrexate (MTX) fate, with implications for local mucosal retention and systemic antifolate efficacy. (B) Commensal β-lactamases and other degradative enzymes can lower antibiotic exposure (e.g., ceftriaxone), promoting resistance and altering therapeutic levels. (C) Microbial immunomodulators (example: Bacteroides fragilis polysaccharide A) increase IL-10 and Treg activation and thereby modulate responses to anti-PD-1 therapy. These mechanistic examples illustrate the direct enzymatic and metabolite-mediated routes discussed in Section 3.

4.2. Microbiome-directed therapeutics in transplantation

Dietary substrates and synbiotic combinations represent an accessible, low-risk approach to selectively fuel beneficial fermentative bacteria and reconstitute metabolite networks that support mucosal and systemic resilience (Andersen et al., 2025; Mathewson et al., 2016; Yue et al., 2024). Small pilot and feasibility studies in allo-HCT recipients have shown that enteral fermentable fibers (e.g., FOS) increase fecal SCFAs and may partially restore community richness; however, these studies have been underpowered for clinical endpoints, and signals for reduced gastrointestinal GVHD remain preliminary and subject to heterogeneity across cohorts (Andermann et al., 2021). Mechanistic studies indicate that fermentable fibers preferentially enrich butyrate-producing and cross-feeding consortia, which reestablish histone acetylation patterns and tighten epithelial barrier integrity in animal models, observations that define a mechanistic pathway in which microbial metabolites alter epithelial biology and thereby influence clinical outcomes; these data argue for scaled, biomarker-driven trials of precisely formulated fermentable substrates in transplant cohorts (Feng et al., 2024; Mathewson et al., 2016; Song et al., 2024; Yue et al., 2024). Although single-strain probiotics (e.g., Lactobacillus rhamnosus GG) show reproducible benefit in murine GVHD models, randomized human trials in allo-HCT have been heterogeneous and in some cases negative, emphasizing that preclinical efficacy does not reliably translate to immunocompromised patients without careful attention to strain choice, timing, and safety monitoring. Clinicians should therefore interpret probiotic efficacy claims cautiously in transplant settings until larger, well-controlled trials establish consistent benefit and safety (Gorshein et al., 2017).

To help translate promise into practice we here explicitly tier evidence by modality in the hematology / transplant context. Prebiotic/synbiotic strategies (for example fermentable fibers such as FOS) have small, mostly single-centre pilots showing feasibility and increases in fecal SCFAs and community richness, but clinical endpoints (GVHD incidence/severity, mortality) remain underpowered in current trials. Probiotics show robust preclinical signals but heterogeneous human outcomes (randomized trials in allo-HCT have not consistently replicated murine benefit), so clinical support is currently inconsistent. Postbiotics / metabolite replacement (for example butyrate analogues) have compelling mechanistic rationale and animal evidence and are now entering early human trials, but clinical efficacy data in transplant populations remain preliminary. Defined bacterial consortia have the strongest randomized human evidence in other clinical contexts (for example VE303 for recurrent C. difficile) and are therefore the most advanced platform technologically, but hematology-specific efficacy and safety data are still limited. Finally, targeted microbial enzyme inhibition (for example selective inhibition of gut β-glucuronidases) has strong mechanistic and preclinical proof-of-concept and early translational data suggesting therapeutic utility to reduce local drug toxicity, yet randomized clinical evidence in hematology is not yet available. We therefore caution that conclusions about patient benefit should be graded by modality: defined consortia and enzyme-targeting approaches are closest to translational proof-of-concept, prebiotics/postbiotics are promising but need larger, biomarker-driven trials, and single-strain probiotics currently show mixed results and require careful strain, timing and context selection (Intestinal, 2023; Andermann et al., 2021; Bhatt et al., 2020; Gorshein et al., 2017; Louie et al., 2023).

Beyond single-strain products, defined bacterial consortia and manufactured microbial therapeutics are advancing in clinical trials and offer the prospect of reproducible metabolic functionality. Importantly, randomized human evidence for defined consortia exists in other clinical contexts (for example VE303 in recurrent C. difficile), illustrating that manufactured consortia are a feasible platform; hematology-specific randomized efficacy and safety data, however, are still needed before routine clinical adoption in transplant populations (Louie et al., 2023). FMT has shown promise in selected post-transplant GVHD case series and early reports as a rescue for refractory dysbiosis, but randomized controlled data in transplant recipients are limited; FMT requires stringent donor screening and careful risk-benefit assessment in immunosuppressed recipients (Faecal, 2021; Gray and DeFilipp, 2023; Weber et al., 2024). An orthogonal precision strategy is selective inhibition of microbial enzymes (for example gut β-glucuronidases) to limit local reactivation of toxic drug metabolites and protect mucosal tissues. Preclinical models demonstrate robust mitigation of irinotecan- and

MPA-associated gut toxicity with selective β-glucuronidase inhibitors, and early translational work supports feasibility; nonetheless, randomized trials in hematology populations are not yet available and safety in profoundly immunosuppressed hosts requires careful evaluation (Lin et al., 2021; Wallace et al., 2010; Wang et al., 2025). Table 3 summarizes the microbiome-directed therapeutic strategies that are most relevant to transplantation and their current evidence and caveats.

Fig. 3 synthesizes mechanistic insights into a clinical roadmap, showing specific microbiome-targeted interventions and biomarkers that could be embedded in trials for allo-HCT, immunotherapy, and chemotherapy to improve efficacy and reduce toxicity.

5. Challenges and limitation

5.1. From 16S to function: multi-omic strategies for mechanistic microbiome research

Researchers increasingly acknowledge that 16S rRNA amplicon sequencing, while useful for rapid taxonomic snapshots, offers limited mechanistic insight because it lacks strain-level resolution, misses non-bacterial members (viruses, fungi), and does not report gene or pathway content required to infer metabolic capacity; these limitations hamper its utility when the goal is to connect microbial communities to chemical mediators of drug response in hematologic patients (Almeida et al., 2021; Leviatan et al., 2022; Liao et al., 2023). Consequently, whole-metagenome shotgun sequencing is now widely adopted when investigators must map microbial gene repertoires (including biosynthetic gene clusters and resistance loci) and resolve organisms at species or strain level, features directly relevant to predicting bacterial drug-modifying activities and immune-modulatory potential (Almeida et al., 2021; Go et al., 2024; Kifle et al., 2024). Shotgun datasets further permit assembly of metagenome-assembled genomes (MAGs) and the taxonomic anchoring of previously uncultured lineages into reference collections, expanding the searchable gene universe and strengthening functional annotations that support testable mechanistic models (Almeida et al., 2021; Leviatan et al., 2022). Because DNA-based surveys reflect genetic potential rather than real-time function, investigators must superimpose metatranscriptomics (to quantify expressed microbial genes) and untargeted metabolomics (to catalog small-molecule outputs) in order to reveal which pathways are actually active under particular host or treatment conditions (Go et al., 2024; Josephs-Spaulding et al., 2025; Pötgens et al., 2024). ndeed, metatranscriptomic analyses can uncover metabolic processes that remain hidden when considering DNA abundance alone, thereby allowing researchers to prioritize biochemical activities with direct bearings on drug biotransformation and immune phenotypes pertinent to blood cancers (Josephs-Spaulding et al., 2025; Xiong et al., 2024). Untargeted metabolomics complements genomic and transcriptomic layers by directly measuring the small molecules produced or altered by microbes and host tissues; integrated holo-omic analyses therefore enable explicit gene-to-metabolite mappings that illuminate biochemical mediators of hematopoiesis, chemo-toxicity, and immunotherapy outcomes (Go et al., 2024; Li et al., 2024; Pötgens et al., 2024). Table 4 lists a practical holo-omic toolkit (modalities and specimen types) that we recommend for mechanistic translational pipelines.

5.2. From correlation to proof: experimental and clinical pipelines for causal inference

Proving causal links between the microbiome and host phenotypes demands tractable experimental platforms that faithfully model human host–microbe relationships; germ-free and gnotobiotic mouse systems are particularly valuable because they allow transplantation of defined bacterial consortia or patient-derived communities into a controlled, microbiota-free host background (Jans and Vereecke, 2025; Kobel et al., 2024; Wang et al., 2022). Humanized and germ-free murine experiments have translated epidemiological signals into mechanistic proof by showing that particular microbes or their metabolic products reprogram systemic immune circuits, modify tumor behavior, and change the efficacy of chemotherapy and immunotherapy (Go et al., 2024; Lei et al., 2025; Wang et al., 2022). Nonetheless, extrapolating rodent-derived causal findings to the clinic is nontrivial: interspecies gaps in immune architecture, physiology, diet, xenobiotic metabolism, and baseline microbial communities, plus treatment-specific factors in hematology (for example, extensive antibiotics, conditioning, and mucosal damage), introduce confounders that limit direct translation (Lei et al., 2025; Pötgens et al., 2024; Wang et al., 2022). Integrative readouts, for example, pairing multi-compartment metabolomics with community profiling, strengthen causal inference by linking specific taxa to their biochemical signatures across body compartments and thereby highlighting the metabolite-mediated axes most likely to affect host biology (Cheng et al., 2022; Go et al., 2024; Josephs-Spaulding et al., 2025). Closing the translational loop necessitates carefully designed human interventions, e.g., donor-derived fecal microbiota transplantation or controlled microbiome-modulation trials, but these approaches raise practical and ethical challenges in immunocompromised hematology patients, demand intensive safety oversight, and commonly produce variable results shaped by donor selection, recipient state, and procedural details (Fecal Microbiota, 2022; Baydoun et al., 2025; Woodworth et al., 2025). To address these limitations, preclinical mechanistic validation should be coupled to early-phase clinical studies imbued with dense biomarker strategies, including metagenomics, metatranscriptomics, and targeted metabolomics, so trials can verify that an intervention elicits the proposed molecular perturbations as well as the clinical benefit, rather than inferring effect from compositional shifts alone (Go et al., 2024; Prosty et al., 2024; Sherwani et al., 2025). Ultimately, a robust causal program in hematologic oncology will rely on iterative loops: multi-omic discovery in patient cohorts to generate

Table 3
Microbiome-directed therapeutic strategies (evidence and caveats).

Intervention	Mechanism / rationale	Evidence summary in hematology / translational context	Ref.
Prebiotic / fermentable fiber	Boost butyrate-producing taxa → restore SCFA pools, strengthen barrier	Pilot transplant trials show increased fecal SCFAs and signals toward reduced GVHD (feasibility data)	(Andersen et al., 2025; Song et al., 2024)
Defined bacterial consortia (e.g., VE303)	Deliver standardized metabolic functionality (butyrate producers / restorative taxa)	Phase II/III work in other fields (e.g., CDI) shows safety and function; consortia are promising for transplant but need hematology-specific trials	(Louie et al., 2023)
Fecal microbiota transplantation (FMT)	Rapid restoration of community richness/ metabolic output	Case series and early trials in post-transplant GVHD show promise but require stringent donor screening and safety oversight	(Gray and DeFilipp, 2023; Weber et al., 2024)
Targeted microbial enzyme inhibitors	Block discrete bacterial enzymes (e.g., GUS) to prevent local reactivation of toxic metabolites	Preclinical models show mitigation of GI toxicity and improved tolerability of chemotherapies; translational enzyme targets under investigation	(Bhatt et al., 2020; Lin et al., 2021; Wallace et al., 2010)
Postbiotics / metabolite replacement (e.g., butyrate analogues)	Replace protective microbial metabolites directly to restore epithelial/immune tone	Pilot data and rationale from animal models; human transplant proof-of-concept data emerging	(Mathewson et al., 2016; Shah et al., 2022)

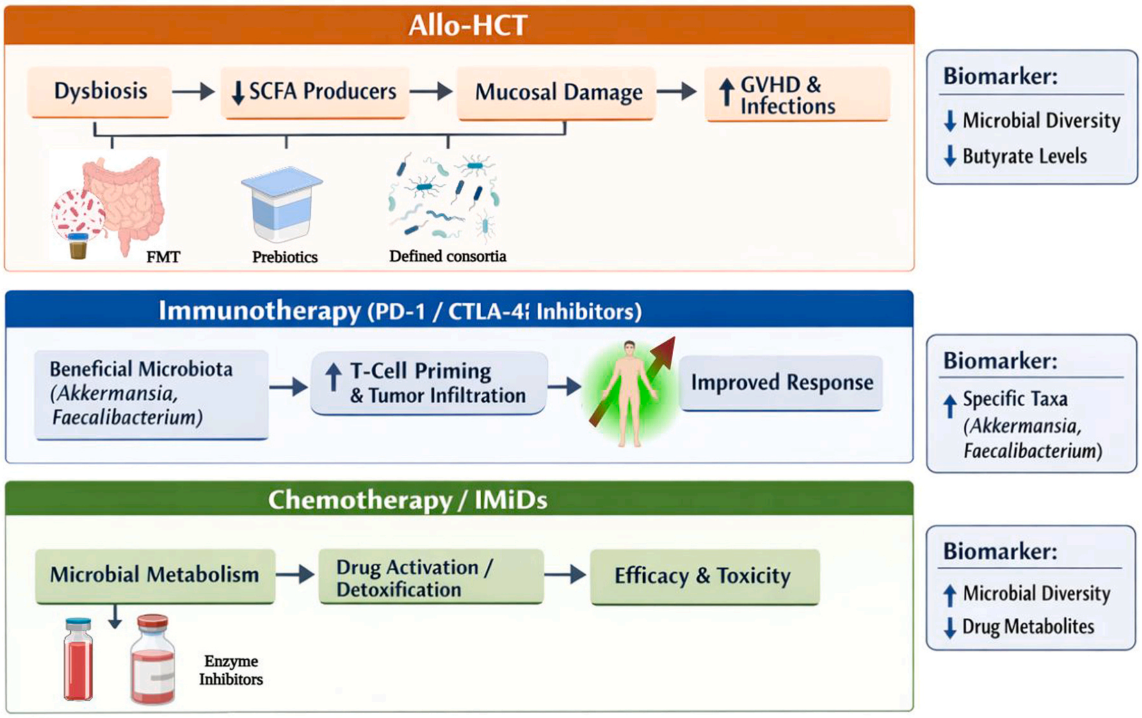

Fig. 3. Translational roadmap: how microbial ecology and metabolites modulate therapy across three clinical contexts (allogeneic HCT, checkpoint immunotherapy, and chemotherapy/IMiDs) and where targeted interventions can be deployed. Top: In allo-HCT, dysbiosis and loss of SCFA producers (↓butyrate) drive mucosal damage, GVHD and infections, interventions include FMT, defined consortia, and metabolite replacement. Middle: Beneficial taxa (e.g., Akkermansia, Faecalibacterium) promote T-cell priming and tumor infiltration, enhancing checkpoint responsiveness. Bottom: Microbial drug metabolism influences activation/detoxification of chemotherapeutics and IMiDs; enzyme inhibitors and microbiome-guided dosing can modulate efficacy and toxicity. The right panels summarize candidate biomarkers to embed in trials (microbial diversity, specific taxa, and drug-metabolite profiles).

hypotheses, targeted falsification in reductionist and humanized models, and mechanistically guided, adequately powered interventional trials that confirm both molecular mechanism and clinical effect (Kobel et al., 2024; Lei et al., 2025; Woodworth et al., 2025). Importantly, many human microbiome–outcome associations in hematology are observed in settings with intense, heterogeneous clinical exposures that can confound inference. Key confounders include timing and spectrum of antimicrobial exposures (which can independently alter both microbiome composition and risk of infection/GVHD), preparative conditioning intensity and radiation, severity and duration of mucositis (which affects barrier function and translocation), nutritional intake and enteral feeding, episode severity of infections (and their treatments), and local hospitalization ecology (ward-level antibiotic stewardship, room environment, and cross-transmission risk). Robust causal inference therefore requires study designs and analyses that explicitly account for these factors (e.g., detailed antibiotic time-series, conditioning stratification, and mediation analyses) (Bansal et al., 2022; Gavriilaki et al., 2020; Rashidi et al., 2023; Shono et al., 2016; Wade and Hall, 2019).

5.3. Holobiont-scale OME-wide integration

Adopting a holobiont viewpoint emphasizes that clinical phenotypes reflect continuous crosstalk among host genetic programs, the resident microbial community and the chemical mediators exchanged between them; as a result, researchers are moving toward integrated 'OME-wide' experiments that simultaneously capture host genomics and transcriptomics together with metagenomes, metatranscriptomes, proteomes and metabolomes (Kobel et al., 2024; Lei et al., 2025; Nyholm et al., 2020). Initial holo-omic studies demonstrate that layering complementary data types materially improves mechanistic insight and prediction compared with single-omic surveys: multi-omic models can, for instance, link a microbial locus that is transcriptionally active to its protein product and to a downstream metabolite that functionally engages a host pathway (Go et al., 2024; Li et al., 2024; Odriozola et al., 2024; Qian et al., 2025). Bringing OME-wide designs into hematologic oncology imposes clear logistical and computational burdens: investigators must harmonize multi-matrix collection protocols, plan longitudinal sampling to resolve treatment-driven dynamics, adopt standardized bioinformatic workflows to mitigate batch and technical artifacts, and deploy advanced inference methods that perform robust feature selection and causal mediation in very high-dimensional datasets (Almeida et al., 2021; Jans and Vereecke, 2025; Prosty et al., 2024; Sherwani et al., 2025; Wu et al., 2025). Recent expansion of reference genome databases and improvements in genome-resolved metagenomic pipelines make it progressively easier to assign microbial genes to biochemical functions, which in turn increases confidence when linking microbial gene expression to metabolite signatures and host biology (Almeida et al., 2021; Kifle et al., 2024; Leviatan et al., 2022). As network-inference, mediation frameworks and multi-omic integrators become more robust, OME-wide studies should reveal microbial metabolites and host circuits that are both mechanistically actionable and clinically actionable biomarkers. Realizing this potential will demand community-wide adoption of reproducible pipelines, thorough methodological transparency, and deliberate inclusion of ethnically and clinically diverse cohorts to secure generalizable results (Lei et al., 2025; Lu et al., 2022; Odriozola et al., 2024; Sherwani et al., 2025).

5.4. Standards and best practices for clinical microbiome/metabolome studies

To increase reproducibility and interpretability in clinical translational pipelines we recommend adopting community standards for study design, sampling, contamination control, analytic QC and

Table 4
Practical holo-omic toolkit for causal translational studies.

Modality / sample	What it measures	Practical note / why include it	Ref.
Whole-metagenome shotgun sequencing (stool / mucosa)	Gene content and strain-level resolution (gene families, biosynthetic clusters)	Necessary to assign drug-modifying enzymes to taxa and to design targeted interventions	(Almeida et al., 2021; Leviatan et al., 2022)
Metatranscriptomics (stool / mucosa)	Actively expressed microbial genes (which pathways are on/off)	Distinguishes genetic potential from active function, essential for mapping enzymes that are actually transcribed under treatment.	(Josephs-Spaulding et al., 2025; Nayak et al., 2021)
Untargeted metabolomics (stool, tissue, plasma)	Small-molecule outputs (SCFAs, bile acids, drug metabolites)	Direct readout of mediators that reach host compartments and bind host receptors, anchors causality.	(Go et al., 2024; Mathewson et al., 2016)
Metaproteomics / targeted enzyme assays (stool/mucosa)	Enzyme abundance and activity (e.g., GUS activity assays)	Verifies that predicted enzymes are present and functional, validates intervention targets (e.g., enzyme inhibitors).	(Bhatt et al., 2020; Wallace et al., 2015)
Spatial / mucosal sampling and paired host omics	Local tissue metabolite gradients and host transcriptional responses	Important in transplantation and mucositis where lumenal vs mucosal vs systemic pools differ.	(Odriozola et al., 2024; Prosty et al., 2024)

reporting. The STORMS checklist provides a comprehensive reporting framework for human microbiome studies (sampling details, negative/positive controls, raw data deposition, analysis pipelines). The Microbiome Quality Control (MBQC) project documents sources of interlaboratory variation and recommends use of standardized controls and replication to quantify batch effects. The Genomic Standards Consortium's MIxS metadata standards enable consistent capture of contextual sample information essential for cross-study harmonization. For computational pipelines and batch-effect mitigation, several methodological reviews provide practical recommendations for benchmarking and normalization. Implementing these standards (STORMS, MBQC, MIxS plus established computational best practices) should be a precondition for prospective holo-omic trials in hematology to ensure reproducible, comparable datasets (Bokulich et al., 2020; Mirzayi et al., 2021; Sinha et al., 2015).

6. Future directions

6.1. Priorities for translational validation and clinical trials

A central near-term priority is to move beyond reproducible observational associations toward prospective, mechanism-embedded interventional trials and rigorous prognostic validation studies in hematologic oncology. For predictive biomarkers, this requires pre-specified model definitions, reporting of discrimination and calibration, and independent (external) validation; for interventions, trials should embed dense molecular endpoints that confirm on-target biological perturbation in addition to clinical benefit (Collins et al., 2015; Steyerberg et al., 2010). Large multicenter cohorts have shown that peri-transplant microbial diversity predicts mortality and graft-versus-host disease risk, establishing a strong rationale for trials that prospectively manipulate microbiota composition or metabolite pools in allogeneic hematopoietic cell transplantation (Mathewson et al., 2016; Peled et al., 2020). We propose embedding metabolome-based risk stratification into HSCT and CAR-T care pathways: baseline and early post-conditioning metabolomic profiles (e.g., short-chain fatty acids, bile acid and tryptophan catabolites, NAD+/redox markers) may stratify patients for intensified infection surveillance, alternative prophylaxis, or metabolite-replacement therapy in order to reduce GVHD, infectious complications, or CAR-T inflammatory toxicity. Early multicenter work and cohort analyses already link metabolite signatures to transplant and CAR-T outcomes and support prospective metabolomic stratification as a realistic near-term objective for biomarker-driven trials (Jalota et al., 2023; Smith et al., 2022).

6.2. Mechanistic, holo-omic pipelines to move from correlation to causation

To infer causality and identify drug-modifying microbial activities, investigators must adopt holo-omic pipelines that pair genome-resolved metagenomics with metatranscriptomics, proteomics and spatially resolved metabolomics across stool, mucosa and systemic compartments. Whole-metagenome assemblies and metatranscriptomes allow assignment of drug-modifying enzymes to specific strains and their transcriptional states, while untargeted metabolomics documents the small molecules that actually reach the host and bind receptors or undergo enterohepatic cycling (Gopalakrishnan et al., 2018; Mathewson et al., 2016). Using these multi-layer data in tandem with gnotobiotic and humanized mouse models will enable falsifiable mechanistic hypotheses, for example mapping specific bacterial β-glucuronidases to mycophenolate reactivation or identifying taxa that convert tacrolimus to inactive metabolites (Bhatt et al., 2020; Guo et al., 2019).

6.3. Precision enzyme targeting and metabolite replacement

A promising precision approach is to inhibit discrete bacterial enzymes that cause local drug reactivation or toxicity while sparing the broader microbiome. Preclinical work targeting bacterial β-glucuronidases has mitigated irinotecan and mycophenolate gut toxicity and enabled therapeutic intensification without wholesale community depletion, illustrating how targeted inhibition of microbial enzymes can reduce gastrointestinal toxicity while preserving systemic drug efficacy (Bhatt et al., 2020; Wallace et al., 2010). Translating these approaches into hematology could mean, for example, co-prescribing validated, non-absorbable bacterial-β-glucuronidase inhibitors during mycophenolate or irinotecan exposure in the peri-transplant period, or developing selective inhibitors aimed at other identified drug-modifying enzymes (e.g., reductases that inactivate specific chemotherapeutics), with early-phase trials using fecal enzyme activity and metabolomic readouts as primary mechanistic endpoints (Louie et al., 2023; Routy et al., 2018). Complementary strategies include development of stable, orally deliverable postbiotics and metabolite analogues that substitute for protective microbial products such as butyrate in settings where butyrogenic communities have been lost after conditioning or antibiotics. Early translational studies of fermentable fiber and defined butyrate-enriching consortia offer proof of concept for metabolite replacement strategies in transplantation and maintenance therapy (Peled et al., 2020; Shah et al., 2022).

6.4. Engineered microbial therapeutics and safety frameworks

Manufactured, defined bacterial consortia offer reproducible metabolic function with standardized safety testing and are now advancing in randomized trials for other indications; adapting these products to hematology will require tailored designs that consider immunosuppression, mucosal integrity and pathogen risk in transplant and neutropenic patients (Gopalakrishnan et al., 2018; Peled et al., 2020). Regulatory pathways must be harmonized so that engineered microbiome therapeutics undergo rigorous donor or strain screening, genomic stability assessment and pharmacokinetic evaluation, and so that trial protocols include robust infection surveillance and criteria for interruption in immunocompromised hosts (Peled et al., 2020).

6.5. Trial design, endpoints and computational standards

Innovative clinical trial designs will accelerate efficient learning; adaptive platform trials that randomize patients to multiple microbiome interventions and use intermediate molecular endpoints such as fecal butyrate, circulating microbial metabolites or measured enzyme activity could rapidly triage promising candidates for larger efficacy studies (Shah et al., 2022). We recommend adopting adaptive, biomarker-driven platform designs (master protocol umbrella/platform trials) that prespecify microbiome/metabolome intermediate endpoints (for example: fecal butyrate > X μmol/g, sustained engraftment of key taxa, or reduction of fecal β-glucuronidase activity) as go/no-go decision rules for arm continuation. Established adaptive oncology platforms (I-SPY2, STAMPEDE, RECOVERY) provide design templates and statistical frameworks that can be adapted for microbiome therapeutics, but require clear a-priori biomarker thresholds, frequent interim molecular monitoring, and regulatory engagement to accept molecular surrogates in early-phase decision making (Angeloni et al., 2025; Park et al., 2023). Equally important are community standards for sample collection, sequencing and metabolomics workflows, and for statistical mediation analyses that account for the high dimensionality of holo-omic data; such harmonization will increase reproducibility and facilitate meta-analysis across institutions (Masetti et al., 2023; Mathewson et al., 2016).

7. Conclusion

Pharmacomicrobiomics offers a concrete opportunity to shrink unexplained variability in drug response and toxicity for patients with hematologic malignancies. Mechanistic and clinical evidence converge to suggest that gut microbes and their metabolites can influence hematopoiesis, drug pharmacokinetics and antitumor immunity; these signals are plausibly linked to clinical outcomes after transplantation, immunotherapy and cellular therapy, but definitive causal proof in humans often remains incomplete and will require mechanism-driven, prospective interventional trials. The scientific imperative now is to translate these insights into rigorous, safety-first interventions that are validated with holo-omic biomarkers and designed for the specific vulnerabilities of hematology patients. Precision approaches that target discrete microbial enzymes, deploy defined microbial consortia, or replace protective metabolites offer the dual advantage of mechanistic specificity and translatability. If the field adopts standardized multiomic pipelines, trials that embed mechanistic endpoints, and harmonized safety frameworks for immunocompromised populations, the gut microbiome can be transformed from a variable confounder into a tractable therapeutic ally that improves the efficacy and tolerability of treatments for blood cancers.

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Data availability

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